

Angular Note 3.3: Advanced Binding

Module: Templates, Data Binding, and Expressions

Topic: Advanced Binding Techniques

Duration: 2-3 hours

Introduction

In Angular applications, data binding is the mechanism that connects the component class with the template. While basic binding techniques like string interpolation and property binding handle most scenarios, Angular provides advanced binding methods for specific use cases. This note explores three specialized binding techniques: Attribute binding, Class binding, and Style binding.

These advanced binding methods give developers precise control over DOM elements, CSS classes, and inline styles, enabling dynamic and responsive user interfaces.

1. Attribute vs Property Binding

1.1 Understanding the Difference

Before diving into attribute binding, it is essential to understand the distinction between HTML **attributes** and DOM **properties**.

HTML Attributes	DOM Properties
Defined in HTML markup	Defined on DOM objects
Initialize DOM properties	Can change dynamically
Values are always strings	Values can be any type
Do not change after initialization	Reflect the current state

Key Point: Not all HTML attributes have corresponding DOM properties, and vice versa. For example, `colspan` is an HTML attribute, but the corresponding DOM property is `colSpan` (note the capital 'S').

1.2 Property Binding (Recap)

Property binding binds a value to a DOM property using square brackets:

```
<img [src]="imageUrl">  
<button [disabled]="isDisabled">Click Me</button>
```

This syntax works when a corresponding DOM property exists.

1.3 Attribute Binding

Attribute binding is required when:

- There is no corresponding DOM property for an HTML attribute
- You need to set ARIA attributes (accessibility)
- You need to work with SVG attributes
- You need to set custom data attributes

Syntax: `[attr.attributeName]="expression"`

1.4 Example: colspan Attribute

The `colspan` attribute specifies how many columns a table cell should span. Let's see both approaches:

Component TypeScript (app.component.ts)

```
import { Component } from '@angular/core';  
  
@Component({  
  selector: 'app-root',  
  templateUrl: './app.component.html',  
  styleUrls: ['./app.component.css']  
})  
export class AppComponent {
```

```
columnSpan = 3;
isFullWidth = true;
}
```

Component Template (app.component.html)

```
<h3>Table with Attribute Binding</h3>
<table border="1" width="100%">
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td [attr.colspan]="columnSpan" style="text-align: center; background-color: #e8f4f8;">
      Employee Information
    </td>
  </tr>
  <tr>
    <td>John Doe</td>
    <td>28</td>
    <td>New York</td>
  </tr>
</table>

<br>

<h3>Property Binding Alternative (colSpan property)</h3>
<table border="1" width="100%">
  <tr>
    <th>Product</th>
    <th>Price</th>
    <th>Stock</th>
  </tr>
  <tr>
    <td [colSpan]="columnSpan" style="text-align: center; background-color: #fff4e8;">
      Product Details
    </td>
  </tr>
  <tr>
    <td>Laptop</td>
    <td>$999</td>
```

```
<td>50</td>
</tr>
</table>
```

Output:

Browser Display:

Two tables will be rendered. In both tables, the second row will have a single cell spanning across all three columns. The first table uses attribute binding `[attr.colspan]`, while the second uses property binding `[colspan]`. Both produce the same visual result.

The cell "Employee Information" will span across Name, Age, and City columns with a light blue background.

The cell "Product Details" will span across Product, Price, and Stock columns with a light orange background.

1.5 ARIA Attribute Example

ARIA (Accessible Rich Internet Applications) attributes improve accessibility. These attributes do not have DOM property equivalents.

Component TypeScript

```
export class AppComponent {
  isExpanded = false;
  currentValue = 60;
  maxValue = 100;

  toggleExpand() {
    this.isExpanded = !this.isExpanded;
  }
}
```

Component Template

```
<button
  (click)="toggleExpand()"
  [attr.aria-expanded]="isExpanded"
  [attr.aria-label]="Toggle content visibility">
  {{ isExpanded ? 'Collapse' : 'Expand' }} Section
</button>

<div *ngIf="isExpanded" style="padding: 10px; background-color: #f0f0f0; margin-top: 10px;">
  <p>This content is now visible!</p>
</div>

<br><br>

<div [attr.role]="progressbar"
  [attr.aria-valuenow]="currentValue"
  [attr.aria-valuemin]="0"
  [attr.aria-valuemax]="maxValue"
  style="width: 300px; height: 30px; background-color: #e0e0e0; border-radius: 5px;">
  <div [style.width.%]="(currentValue / maxValue) * 100"
    style="height: 100%; background-color: #4CAF50; border-radius: 5px; transition: width
0.3s;">
  </div>
</div>
<p>Progress: {{ currentValue }}%</p>
```

Output:

Browser Display:

A button labeled "Expand Section" appears. When clicked, it changes to "Collapse Section" and displays a gray box with text "This content is now visible!"

Below that, a progress bar shows 60% completion in green. Screen readers will announce the ARIA attributes, making the interface accessible to users with disabilities.

When to Use Attribute Binding:

- Setting `colspan` , `rowspan` in tables
- ARIA attributes (`aria-label` , `aria-expanded` , etc.)
- SVG attributes
- Custom data attributes (`data-*`)

2. Class Binding

2.1 Introduction to Class Binding

Class binding allows you to dynamically add or remove CSS classes from an element based on component properties. This is particularly useful for:

- Toggle states (active/inactive)
- Conditional styling
- User interactions (hover, click states)
- Form validation states

2.2 Single Class Binding

Syntax: `[class.className]="condition"`

The class is applied when the condition evaluates to `true` , and removed when it evaluates to `false` .

Component TypeScript

```
export class AppComponent {
  isActive = true;
  isHighlighted = false;
  hasError = true;
  isDisabled = false;

  toggleActive() {
    this.isActive = !this.isActive;
  }
}
```

```
toggleHighlight() {  
  this.isHighlighted = !this.isHighlighted;  
}  
}
```

Component CSS (app.component.css)

```
.active {  
  background-color: #4CAF50;  
  color: white;  
  font-weight: bold;  
  padding: 10px;  
  border-radius: 5px;  
}  
  
.highlight {  
  border: 3px solid #ff9800;  
  box-shadow: 0 0 10px rgba(255, 152, 0, 0.5);  
}  
  
.error {  
  color: red;  
  border: 2px solid red;  
  padding: 10px;  
  background-color: #ffe6e6;  
}  
  
.disabled {  
  opacity: 0.5;  
  cursor: not-allowed;  
  background-color: #cccccc;  
}  
  
.status-box {  
  padding: 15px;  
  margin: 10px 0;  
  border-radius: 5px;  
  transition: all 0.3s ease;  
}
```

Component Template

```
<h3>Single Class Binding Example</h3>

<div class="status-box" [class.active]="isActive">
  This element is {{ isActive ? 'ACTIVE' : 'INACTIVE' }}
</div>

<button (click)="toggleActive()">Toggle Active State</button>

<br><br>

<div class="status-box" [class.highlight]="isHighlighted">
  This element is {{ isHighlighted ? 'HIGHLIGHTED' : 'NORMAL' }}
</div>

<button (click)="toggleHighlight()">Toggle Highlight</button>

<br><br>

<div [class.error]="hasError">
  ⚠ Error: This field is required!
</div>
```

Output:

Browser Display:

Three sections appear:

1. A green box with white text saying "This element is ACTIVE" (because isActive is true). Below it, a button labeled "Toggle Active State"
2. A normal gray box saying "This element is NORMAL". Below it, a button labeled "Toggle Highlight"
3. A red-bordered box with light red background showing "⚠ Error: This field is required!" in red text

When you click "Toggle Active State", the first box switches between green (active) and gray (inactive).

When you click "Toggle Highlight", the second box gets an orange border with shadow effect.

2.3 Multiple Class Binding

You can bind to the `class` attribute directly to set multiple classes at once.

Using String

```
export class AppComponent {  
  statusClasses = 'active highlight';  
}
```

```
<div [class]="statusClasses">Multiple classes from string</div>
```

Using Array

```
export class AppComponent {  
  classList = ['active', 'highlight', 'large'];  
}
```

```
<div [class]="classList">Multiple classes from array</div>
```

Using Object (Most Flexible)

```
export class AppComponent {  
  isActive = true;  
  hasError = false;  
  isLarge = true;  
  
  classObject = {  
    'active': this.isActive,  
    'error': this.hasError,
```

```
'large-text': this.isLarge  
};  
}
```

```
<div [class]="classObject">Multiple classes from object</div>
```

2.4 Practical Example: Navigation Menu

Component TypeScript

```
export class AppComponent {  
  currentPage = 'home';  
  
  navigateTo(page: string) {  
    this.currentPage = page;  
  }  
  
  isActivePage(page: string): boolean {  
    return this.currentPage === page;  
  }  
}
```

Component CSS

```
.nav-menu {  
  display: flex;  
  gap: 5px;  
  background-color: #333;  
  padding: 10px;  
  border-radius: 5px;  
}  
  
.nav-item {  
  padding: 10px 20px;  
  color: white;  
  background-color: #555;  
  cursor: pointer;  
  border-radius: 3px;  
  transition: background-color 0.3s;  
}
```

```
.nav-item:hover {  
  background-color: #777;  
}  
  
.nav-item.active {  
  background-color: #4CAF50;  
  font-weight: bold;  
}  
  
.content-area {  
  margin-top: 20px;  
  padding: 20px;  
  background-color: #f5f5f5;  
  border-radius: 5px;  
  min-height: 100px;  
}
```

Component Template

```
<div class="nav-menu">  
  <div class="nav-item"  
    [class.active]="isActivePage('home')"  
    (click)="navigateTo('home')">  
    Home  
  </div>  
  <div class="nav-item"  
    [class.active]="isActivePage('about')"  
    (click)="navigateTo('about')">  
    About  
  </div>  
  <div class="nav-item"  
    [class.active]="isActivePage('services')"  
    (click)="navigateTo('services')">  
    Services  
  </div>  
  <div class="nav-item"  
    [class.active]="isActivePage('contact')"  
    (click)="navigateTo('contact')">  
    Contact  
  </div>  
</div>  
  
<div class="content-area">
```

```
<h2>Current Page: {{ currentPage | uppercase }}</h2>
<p>This is the content for the {{ currentPage }} page.</p>
</div>
```

Output:

Browser Display:

A dark gray navigation bar appears with four items: Home, About, Services, Contact.

The "Home" button is highlighted in green with bold text (active state).

Below the menu, a light gray content area shows "Current Page: HOME" and "This is the content for the home page."

When you click on any menu item, it becomes green (active), and the previous active item returns to gray. The content area updates to show the selected page name.

3. Style Binding

3.1 Introduction to Style Binding

Style binding allows you to set inline CSS styles dynamically based on component data. While CSS classes are preferred for most styling scenarios, style binding is useful when:

- Styles depend on runtime calculations
- You need to set dynamic values (colors, dimensions, positions)
- Working with animations or transitions
- User preferences affect styling

3.2 Single Style Property Binding

Syntax: `[style.propertyName]="expression"`

You can optionally specify a unit: `[style.width.px]="widthValue"`

Component TypeScript

```
export class AppComponent {
  textColor = 'red';
  fontSize = 20;
  backgroundColor = '#e8f5e9';
  borderWidth = 3;
  isLarge = false;

  changeColor(color: string) {
    this.textColor = color;
  }

  increaseFontSize() {
    this.fontSize += 2;
  }

  decreaseFontSize() {
    if (this.fontSize > 10) {
      this.fontSize -= 2;
    }
  }
}
```

Component Template

```
<h3>Single Style Property Binding</h3>

<div [style.color]="textColor"
    [style.font-size.px]="fontSize"
    [style.background-color]="backgroundColor"
    [style.padding.px]="15"
    [style.border-radius.px]="5">
  This text has dynamic styling!
  Color: {{ textColor }}, Font Size: {{ fontSize }}px
</div>

<br>

<button (click)="changeColor('red')">Red</button>
<button (click)="changeColor('blue')">Blue</button>
```

```
<button (click)="changeColor('green')">Green</button>
<button (click)="changeColor('#ff6b6b')">Pink</button>

<br><br>

<button (click)="increaseFontSize()">Increase Font Size</button>
<button (click)="decreaseFontSize()">Decrease Font Size</button>
```

Output:

Browser Display:

A light green box with red text at 20px size displays: "This text has dynamic styling! Color: red, Font Size: 20px"

Below are four color buttons (Red, Blue, Green, Pink) and two font size buttons (Increase/Decrease).

Clicking "Blue" changes the text color to blue and updates the displayed color value.

Clicking "Increase Font Size" makes the text larger and updates the displayed size. The minimum font size is 10px.

3.3 Multiple Style Properties

You can bind multiple styles using an object.

Component TypeScript

```
export class AppComponent {
  boxStyles = {
    'width': '200px',
    'height': '200px',
    'background-color': '#3f51b5',
    'border-radius': '10px',
    'display': 'flex',
    'align-items': 'center',
    'justify-content': 'center',
    'color': 'white',
    'font-size': '18px',
```

```
'font-weight': 'bold',  
'box-shadow': '0 4px 8px rgba(0,0,0,0.2)'  
};  
  
updateBoxColor(color: string) {  
  this.boxStyles['background-color'] = color;  
}  
}
```

Component Template

```
<div [ngStyle]="boxStyles">  
  Styled Box  
</div>  
  
<br>  
  
<button (click)="updateBoxColor('#e91e63')">Pink</button>  
<button (click)="updateBoxColor('#4caf50')">Green</button>  
<button (click)="updateBoxColor('#ff9800')">Orange</button>
```

Output:

Browser Display:

A blue square box (200x200px) with rounded corners and shadow appears. The text "Styled Box" is centered in white.

Three buttons below allow changing the box color to pink, green, or orange.

Clicking any button immediately changes the box background color while maintaining all other styles.

3.4 Practical Example: Progress Bar

Component TypeScript

```
export class AppComponent {
  progress = 0;
  maxProgress = 100;

  get progressColor(): string {
    if (this.progress < 30) return '#f44336'; // Red
    if (this.progress < 70) return '#ff9800'; // Orange
    return '#4caf50'; // Green
  }

  get progressWidth(): string {
    return this.progress + '%';
  }

  increaseProgress() {
    if (this.progress < this.maxProgress) {
      this.progress += 10;
    }
  }

  decreaseProgress() {
    if (this.progress > 0) {
      this.progress -= 10;
    }
  }

  resetProgress() {
    this.progress = 0;
  }
}
```

Component Template

```
<h3>Dynamic Progress Bar</h3>

<div style="width: 100%; background-color: #e0e0e0; height: 40px; border-radius: 20px;
overflow: hidden;">
  <div [style.width]="progressWidth"
    [style.background-color]="progressColor">
```

```
        style="height: 100%; transition: all 0.3s ease; display: flex; align-items: center;
justify-content: center; color: white; font-weight: bold;">
        {{ progress }}%
    </div>
</div>

<br>

<button (click)="increaseProgress()">+10%</button>
<button (click)="decreaseProgress()">-10%</button>
<button (click)="resetProgress()">Reset</button>

<p>
    Status:
    <span [style.color]="progressColor" [style.font-weight]="bold">
        {{ progress < 30 ? 'Low' : progress < 70 ? 'Medium' : 'High' }}
    </span>
</p>
```

Output:

Browser Display:

A progress bar with gray background shows 0% (empty). Below it are three buttons: "+10%", "-10%", and "Reset".

Status shows "Low" in red.

Clicking "+10%" five times:

- At 20%: Red bar shows "20%" in white text, Status: "Low" (red)
- At 40%: Orange bar shows "40%", Status: "Medium" (orange)
- At 80%: Green bar shows "80%", Status: "High" (green)

The color changes smoothly due to the CSS transition.

3.5 Conditional Style Binding

Combining style binding with conditional logic creates powerful dynamic interfaces.

Component TypeScript

```
export class AppComponent {
  temperature = 25; // Celsius
  humidity = 60; // Percentage

  get temperatureColor(): string {
    if (this.temperature < 15) return '#2196f3'; // Blue (Cold)
    if (this.temperature < 25) return '#4caf50'; // Green (Comfortable)
    if (this.temperature < 35) return '#ff9800'; // Orange (Warm)
    return '#f44336'; // Red (Hot)
  }

  get humidityBackground(): string {
    if (this.humidity < 30) return '#fff9c4'; // Low
    if (this.humidity < 60) return '#c8e6c9'; // Comfortable
    return '#bbdefb'; // High
  }

  increaseTemp() {
    this.temperature += 5;
  }

  decreaseTemp() {
    this.temperature -= 5;
  }
}
```

Component Template

```
<h3>Weather Monitor</h3>

<div [style.background-color]="humidityBackground"
  style="padding: 20px; border-radius: 10px; margin: 10px 0;">

  <div [style.color]="temperatureColor"
    [style.font-size.px]="48"
    [style.font-weight]=" 'bold' "
    style="text-align: center;">
    {{ temperature }}°C
  </div>

  <p style="text-align: center; margin: 10px 0;">
    Humidity: {{ humidity }}%
```

```
</p>

<div style="text-align: center;">
  <button (click)="increaseTemp()">⬆ Increase Temp</button>
  <button (click)="decreaseTemp()">⬇ Decrease Temp</button>
</div>

</div>
```

Output:

Browser Display:

A light green box displays "25°C" in large green text (comfortable temperature), centered.

Below it shows "Humidity: 60%"

Two buttons allow temperature adjustment.

When temperature reaches 35°C or above, the number turns red (hot).

When temperature drops below 15°C, the number turns blue (cold).

The background color changes based on humidity levels, providing visual feedback for environmental conditions.

4. Combining Advanced Binding Techniques

4.1 Real-World Example: Card Component

Let's create a comprehensive example that uses all three advanced binding techniques together.

Component TypeScript

```
export class AppComponent {  
  cards = [  
    { id: 1, title: 'Task 1', status: 'pending', priority: 'high' },  
    { id: 2, title: 'Task 2', status: 'completed', priority: 'low' },  
    { id: 3, title: 'Task 3', status: 'in-progress', priority: 'medium' },  
    { id: 4, title: 'Task 4', status: 'pending', priority: 'high' }  
  ];  
  
  getStatusColor(status: string): string {  
    const colors: any = {  
      'pending': '#ff9800',  
      'in-progress': '#2196f3',  
      'completed': '#4caf50'  
    };  
    return colors[status] || '#9e9e9e';  
  }  
  
  getPriorityBorder(priority: string): string {  
    const borders: any = {  
      'high': '3px solid #f44336',  
      'medium': '3px solid #ff9800',  
      'low': '3px solid #4caf50'  
    };  
    return borders[priority] || '1px solid #ccc';  
  }  
  
  isCompleted(status: string): boolean {  
    return status === 'completed';  
  }  
  
  toggleStatus(card: any) {  
    if (card.status === 'pending') {  
      card.status = 'in-progress';  
    } else if (card.status === 'in-progress') {  
      card.status = 'completed';  
    } else {  
      card.status = 'pending';  
    }  
  }  
}
```

Component CSS

```
.card-container {  
  display: flex;  
  gap: 15px;  
  flex-wrap: wrap;  
  margin: 20px 0;  
}  
  
.card {  
  width: 200px;  
  padding: 15px;  
  border-radius: 8px;  
  background-color: white;  
  box-shadow: 0 2px 4px rgba(0,0,0,0.1);  
  transition: all 0.3s ease;  
  cursor: pointer;  
}  
  
.card:hover {  
  transform: translateY(-5px);  
  box-shadow: 0 4px 8px rgba(0,0,0,0.2);  
}  
  
.card.completed {  
  opacity: 0.7;  
  background-color: #f5f5f5;  
}  
  
.card-title {  
  font-size: 18px;  
  font-weight: bold;  
  margin-bottom: 10px;  
}  
  
.card-status {  
  display: inline-block;  
  padding: 5px 10px;  
  border-radius: 15px;  
  font-size: 12px;  
  color: white;  
  font-weight: bold;  
  text-transform: uppercase;  
}
```

Component Template

```
<h3>Task Management Dashboard</h3>

<div class="card-container">
  <div *ngFor="let card of cards"
    class="card"
    [class.completed]="isCompleted(card.status)"
    [style.border]="getPriorityBorder(card.priority)"
    [attr.data-card-id]="card.id"
    (click)="toggleStatus(card)">

    <div class="card-title">{{ card.title }}</div>

    <div class="card-status"
      [style.background-color]="getStatusColor(card.status)">
      {{ card.status }}
    </div>

    <p style="margin-top: 10px; font-size: 12px; color: #666;">
      Priority:
      <span [style.color]="card.priority === 'high' ? '#f44336' :
        card.priority === 'medium' ? '#ff9800' : '#4caf50'"
        [style.font-weight]="card.priority === 'high' ? 'bold' : 'normal'">
        {{ card.priority | uppercase }}
      </span>
    </p>
  </div>
</div>

<p style="font-size: 12px; color: #666; font-style: italic;">
  💡 Tip: Click on any card to change its status
</p>
```

Output:

Browser Display:

Four task cards appear in a row:

1. **Task 1:** White card with thick red border (high priority). Orange badge says "PENDING". Priority shows in bold red.

2. **Task 2:** Grayed out card with green border (low priority). Green badge says "COMPLETED". Priority shows in green.

3. **Task 3:** White card with orange border (medium priority). Blue badge says "IN-PROGRESS". Priority shows in orange.

4. **Task 4:** White card with red border (high priority). Orange badge says "PENDING". Priority shows in bold red.

Clicking Task 1 changes its status: pending → in-progress (blue badge) → completed (grayed card, green badge) → pending (cycles back).

Hovering over any card lifts it slightly with enhanced shadow.

5. Best Practices and Guidelines

5.1 When to Use Each Binding Type

Binding Type	Use Case	Example
Attribute Binding	No corresponding DOM property exists	<code>[attr.colspan]</code> , <code>[attr.aria-label]</code>
Class Binding	Toggle CSS classes conditionally	<code>[class.active]="isActive"</code>
Style Binding	Dynamic inline styles with runtime values	<code>[style.width.px]="width"</code>

5.2 Performance Considerations

- Prefer CSS classes over inline styles when possible for better performance and maintainability
- Use class binding for predefined styles that toggle on/off
- Use style binding only when values are calculated at runtime
- Avoid complex expressions in template bindings; move logic to component methods

- Use `ngStyle` and `ngClass` directives for multiple properties

5.3 Common Mistakes to Avoid

1. Confusing attributes and properties:

```
<!-- ✗ Wrong: colspan is not a property -->
<td [colspan]="3"></td>

<!-- ✓ Correct: Use attribute binding -->
<td [attr.colspan]="3"></td>

<!-- ✓ Also correct: Use property binding with correct property name -->
<td [colSpan]="3"></td>
```

2. Overusing inline styles:

```
<!-- ✗ Not recommended: Too many inline styles -->
<div [style.color]='red'
      [style.background]='blue'
      [style.padding]='10px'
      [style.margin]='5px'></div>

<!-- ✓ Better: Use CSS class -->
<div [class.error-box]="true"></div>
```

3. Forgetting units in style binding:

```
<!-- ✗ Wrong: No unit specified -->
<div [style.width]="200"></div>

<!-- ✓ Correct: Unit specified -->
<div [style.width.px]="200"></div>
<div [style.width]='200px'></div>
```

5.4 Accessibility Considerations

- Always use `[attr.aria-*)` attributes for screen reader support

- Ensure color is not the only indicator of state (use text, icons, or patterns too)
- Maintain sufficient color contrast ratios (WCAG guidelines)
- Use semantic HTML with appropriate ARIA roles

Summary

Advanced binding techniques in Angular provide fine-grained control over DOM elements:

1. Attribute Binding (`[attr.name]`):

- Used when no DOM property exists
- Essential for ARIA attributes, SVG, and table attributes
- Syntax: `[attr.colspan]="value"`

2. Class Binding (`[class.name]`):

- Conditionally apply CSS classes
- Preferred method for toggling predefined styles
- Syntax: `[class.active]="isActive"`

3. Style Binding (`[style.property]`):

- Set inline CSS styles dynamically
- Useful for runtime-calculated values
- Syntax: `[style.color]='red'` or `[style.width.px]="100"`

These techniques work together to create dynamic, responsive, and accessible user interfaces.

Understanding when and how to use each binding type is crucial for writing clean, maintainable Angular applications.

Practice Exercises

1. **Exercise 1:** Create a data table with dynamic column spanning based on user selection.

2. **Exercise 2:** Build a theme switcher that toggles between light and dark modes using class binding.
 3. **Exercise 3:** Implement a dynamic sizing control where users can adjust element width and height using style binding.
 4. **Exercise 4:** Create an accessible form with proper ARIA attributes using attribute binding.
 5. **Exercise 5:** Build a status dashboard that combines all three binding types to display system metrics with color-coded indicators.
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End of Angular Note 3.3: Advanced Binding

